Java

Class & object

```java
public class oops {
    private int data;
    String name;
    oops(){
        System.out.println("constructor called");
    }
    oops(int value , String name){
        data = value;
        this.name = name;
    }
    oops(oops original){
        this.data = original.data;
        this.name = original.name;
    }

    void setName(String name){
        this.name = name;
    }

    void setData(int data){
        this.data = data;
    }
    // class => blueprint
    // object => instance of a class
    // variable ( fields)
    // function (method)
    void displayInfo(){
        System.out.println(data + " updated value "+name );
    }
    public static void main(String[] args) {

        // oops obj = new oops();
        // obj.data = 60;
        // obj.displayInfo();

        // oops obj1 = new oops(100 , "FIIT");
        // // obj1.data = 20;
        // obj1.displayInfo();
        // oops copyObj = new oops(obj1); // copy constructor
        // copyObj.displayInfo();

        // encapsulation => setter and getter

        oops encap = new oops();
```

```
        encap.setData(40);
        encap.setName("FIIT");
        encap.displayInfo();

        // constructor => special method or function
        // call itself => when object created
        // (){

        // }
        // default constructor (no  argument constructor)
        // parameterized constructor
        // copy constructor
    }

}
// cd => change directory
// ls => list

// javac oops.java => oops.class
```

Tasks:

- Create a `Student` class with fields `id`, `name`, and `marks`. Add constructors and a `display()` method.
- Write a copy constructor for `Student` that duplicates another student object.
- Add a method `calculateGrade()` that returns grade based on marks.
- Create multiple objects and demonstrate how each maintains its own state.